

Does the Williams Cycle apply to Mandarin Chinese?

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Abstract. This paper addresses a theoretical tension in accounting for the correlation between the landing site and the locality domain of movement—whether to reduce it to the $A/\bar{A}$ -distinction (*Ban on Improper Movement*) or to hard-wire it to the grammar as the *Williams Cycle* (WC), which has recently been claimed to apply to Mandarin Chinese by Yan & Meadows (2025). By refuting their arguments, we argue that the WC does not apply to Mandarin and therefore cannot be universal. Specifically, we show that the apparent WC effects in Mandarin either follow from independently attested $A/\bar{A}$ -distinction in the language, or simply do not hold due to incomplete generalization. The Mandarin case highlights the importance of cross-linguistic variation in adjudicating between competing theoretical accounts.

Keywords. height-locality correlation; $A/\bar{A}$ -distinction; Williams Cycle; movement

1. Introduction. As long observed, there is an apparent correlation between the *height* (i.e., landing site) and the *locality domain* of different movement dependencies (Postal 1971; Chomsky 1973; Williams 1974, *i.a.*). To exemplify, raising to subject in English targets Spec,TP and may cross a nonfinite TP but not a finite CP (i.e., no *hyperraising*, Ura 1994), as in (1). By contrast, *wh*-movement targets Spec,CP and may cross both a nonfinite TP and a finite CP, as in (2).

(1) Raising to subject in English cannot cross finite CP

- a. $[_{TP} \text{ Alex seems } [_{TP} \text{ to } __ \text{ dislike New Haven pizzas}]]$.
↑ ✓ A-movement
- b. $*[_{TP} \text{ Alex seems } [_{CP} \text{ that } __ \text{ dislikes New Haven pizzas}]]$. (No hyperraising)
↑ ✗ A-movement

(2) Wh-movement in English can cross finite CP

- a. $[_{CP} \text{ What pizzas did } [_{TP} \text{ Alex claim } [_{TP} \text{ to dislike } __]]]?$
↑ ✓ $\bar{A}$ -movement
- b. $[_{CP} \text{ What pizzas did } [_{TP} \text{ Alex claim } [_{CP} \text{ that she dislikes } __]]]?$
↑ ✓ $\bar{A}$ -movement

There have been two different routes to theorize contrasts like the one in (1)–(2). One dominant route by Chomsky (1973, 1977) is to distinguish movement by two types: A-movement that lands on A-positions such as Spec,TP, and $\bar{A}$ -movement that lands on $\bar{A}$ -positions such as Spec,CP. The locality restriction follows from the *Ban on Improper Movement* stated in (3): A-movement, such as raising to subject, cannot proceed via Spec,CP.

(3) Ban on Improper Movement (after Chomsky 1973)

An element may not be moved from an $\bar{A}$ -position to an A-position.

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The A/ $\bar{A}$ -distinction approach has been extremely successful in capturing not just locality but also other movement properties including weak crossover and reconstruction patterns (see overviews in Richards 2014; Safir 2019). Recent lines of proposals attempt to reduce (3) to a primitive distinction between movement types, such as Obata & Epstein (2011)’s featural approach (see also van Urk 2015; Chen 2023, 2025b; Lohninger et al. 2022; Lohninger & Yip 2023; Lohninger 2025; Branen & Erlewine 2024, *i.a.*) where A-features on $\bar{A}$ -moved elements are not visible to a higher probe; and Safir (2019)’s categorial approach (see also Hewett 2025) where $\bar{A}$ -moved elements have an extra layer QP (Cable 2007) that shields the A-features from a higher probe. Importantly, the apparent height-locality correlation is not hard-wired into the grammar but only arises if certain higher heads like C always host $\bar{A}$ -probes in a language (e.g., English, but see Greenson 2025). Given that heads may lexically bundle different probing features across languages, this correlation is not expected to hold universally. And this is borne out by the fact that, while English does not, there are other languages that permit hyperraising (Fong 2019; Lohninger 2025), including Mandarin Chinese, which is the focus of this paper.

Another route, proposed by Williams (1974, 2003, 2013), encodes the height-locality correlation directly in the grammar, as the *Williams Cycle* (WC) stated in (4).

(4) Williams Cycle (after Williams 1974, 2003, 2013)

Movement to Spec,XP cannot proceed from Spec,YP or across YP, where Y is higher than X in the functional sequence.

WC effects have received recent cross-linguistic support (e.g., Swahili, Meadows 2024; Georgian, Bondarenko 2024; Hungarian, Egressy 2025) and have been argued to extend to non-movement phenomena (e.g., case in Finnish, Poole 2022; sequence of tense in Hungarian, Egressy 2026b). Proposals that attempt to derive the WC effects include the Level Embedding approach by Williams (2003) himself (see also Meadows 2024), where locality follows from structure building (see more discussion in §2), as well as Keine (2020)’s Horizon approach (see also Kobayashi 2020; Messick & Raghotham 2024), where each probe is specified with its locality domain.

The central question in this paper is whether the WC applies to Mandarin Chinese. Recently, Yan & Meadows (2025) (henceforth *YM25*; superseding Meadows & Yan 2025) argue that it does, drawing on object topicalization and Verb Copying constructions as illustrated in (5)–(6).

(5) Object topicalization in Mandarin

[**Zhe-ben shu**] Lisi bu hui kan ____.
this-CL book Lisi not will read
‘This book, Lisi will not read.’

(6) Verb Copying constructions in Mandarin

[**Kan zhe-ben shu**] Lisi kan-de hen kuai.
read this-CL book Lisi read-DE very fast
‘Read this book, Lisi reads (it) fast.’

However, YM25’s claim is at odds with earlier work showing that Mandarin permits *hyper-raising* to subjects across a finite CP (Chen 2023, 2025b; Lee & Yip 2024; Chou 2026), which constitutes a direct counterexample to the WC. To resolve this tension, one must either incorporate hyperraising into the WC framework or derive the apparent WC effects using independently motivated mechanisms in the grammar.

This paper pursues the second approach. We reexamine the arguments in YM25 and show that the apparent WC effects in Mandarin either reduce to A/ $\bar{A}$ -distinction of movement with diagnosable distribution of pure [$\bar{A}$] versus composite [A/ $\bar{A}$] probes, or simply do not hold due

to incomplete generalization. Therefore, the WC does *not* apply to Mandarin. This conclusion implies that the WC cannot be universal and is, at best, parameterized across languages, if not untenable altogether. More generally, it raises the question of how cross-linguistic variation should be integrated into the WC framework, especially given that research on the A/ $\bar{A}$ -distinction has already produced more fine-grained and empirically grounded accounts (e.g., Lohninger 2025 and references therein).

The paper is organized as follows. §2 discusses YM25’s claim for the WC in Mandarin. §3 presents arguments against their claim, from hyperraising, object topicalization, and VC constructions. §4 concludes.

2. Yan & Meadows (2025)’s apparent Williams Cycle effects in Mandarin. YM25 argue that Mandarin exhibits WC effects in two movement dependencies: *object topicalization* (which they term *object shift*) and *Verb Copying constructions* (which they term *verb doubling*). Both dependencies may target either a high, TP/IP-external (henceforth *external*) position, as in (5)–(6) above, or a low, TP/IP-internal (henceforth *internal*) position, as in (7)–(8) below.

- | | |
|---|--|
| <p>(7) <u>Internal object topicalization</u></p> <p>Lisi [zhe-ben shu] bu hui kan _.</p> <p>Lisi this-CL book not will read</p> <p>‘Lisi, this book, will not read.’</p> | <p>(8) <u>Internal VC constructions</u></p> <p>Lisi [kan zhe-ben shu] kan-de hen kuai.</p> <p>Lisi read this-CL book read-DE very fast</p> <p>‘Lisi reads this book reads (it) fast.’</p> |
|---|--|

YM25 assume that external and internal object topicalization and VC constructions have different landing sites along the functional sequence $C \succ T \succ (M^{Deo}) \succ F \succ Voice \succ V$. Specifically, external object topicalization and VC target Spec,CP, internal object topicalization targets Spec,TP, and internal VC targets Spec,FP,¹ where CPs, TPs, and FPs are selected by ‘know’-type, ‘force’-type (effectively object control), and ‘try’-type (effectively subject control) verbs, following Huang (2022) (see also N. Huang 2018; He 2024; Liu & Yip 2026 for relevant discussions).

The difference in the locality profiles of internal and external object topicalization is well studied in the literature (Qu 1994; Ting 1995; Shyu 1995; Kuo 2009; Chen 2023, 2025a). YM25 make the same observation that internal object topicalization can only cross nonfinite clauses taken by ‘force’-type and ‘try’-type verbs but not finite CPs taken by ‘know’-type verbs, whereas external object topicalization can cross both nonfinite clauses and finite CPs, as illustrated in (9).

- (9) a. Object topicalization with ‘know’-type verbs
 {**Zhe-ben shu_i**} meiyou-ren {***zhe-ben shu_i**} xiangxin/zhidao [_{CP} Lisi hui kan __i].
 this-CL book no-person this-CL book believe/know Lisi will read
 ‘{This book}, no one {*this book}, believes/knows that Lisi will read (it).’
- b. Object topicalization with ‘force’-type and ‘try’-type verbs
 {**Zhe-ben shu_i**} meiyou-ren_j {**zhe-ben shu_i**} dasuan/changshi [_{FP} PRO_j bipo/mingling
 this-CL book no-person this-CL book plan/try force/order
 Lisi_k [_{TP} PRO_k kan __i]].
 Lisi read
 ‘{This book}, no one {this book}, planned/tried to force/order Lisi to read (it).’

¹ They do not make clear what FP is. For concreteness, we understand it as (outer) AspP in the sense of Liu & Yip (2026) and references therein.

Internal VC also cannot cross finite CPs, as observed by Huang (2022:35). YM25 further note that internal VC also cannot cross nonfinite TPs selected by ‘force’-type verbs, a restriction not discussed by Huang in relation to VC. By contrast, external VC can cross both nonfinite clauses and finite CPs. The relevant data are illustrated in (10).

(10) a. VC with ‘know’-type verbs

{**Kan zhe-ben shu**} Zhangsan {***kan zhe-ben shu**} xiangxin/zhidao [_{CP} Lisi hui
read this-CL book Zhangsan read this-CL book believe/know Lisi will
kan-de hen kuai].
kan-DE very fast
‘{Reading this book}, Zhangsan {*reading this book} believes/knows that Lisi will
read (it) fast.’

b. VC with ‘force’-type verbs

{**Kan zhe-ben shu**} Zhangsan {***kan zhe-ben shu**} bipo/mingling Lisi_j [_{TP} PRO_j
read this-CL book Zhangsan read this-CL book force/order Lisi
kan-de hen kuai].
read-DE very fast
‘{Reading this book}, Zhangsan {*reading this book} forced/ordered Lisi to read (it)
fast.’

c. VC with ‘try’-type verbs

{**Kan zhe-ben shu**} Zhangsan {**kan zhe-ben shu**} dasuan/changshi [_{FP} PRO_j
read this-CL book Zhangsan read this-CL book plan/try
kan-de hen kuai].
read-DE very fast
‘{Reading this book}, Zhangsan {reading this book} planned/tried to read (it) fast.’

For YM25, (9)–(10) support a height-locality correlation: external object topicalization and VC, which target Spec,CP, can cross CPs; internal object topicalization, which targets Spec,TP, can only cross TPs and FPs; internal VC, which target Spec,FP, can only cross FPs. To derive these patterns, YM25 adopt the *Level Embedding Approach* (Williams 2003; Meadows 2024), under which embedded and matrix clauses are constructed in parallel in separate workspaces and merged only later. They also assume the *Strict Cycle Condition*, which requires that movement occurs within the cycle in which the relevant structure is built. On this view, YP is built later than XP if Y is higher than X in the functional sequence. These two components together predict the WC in (4): for XP to move across an embedded YP into the matrix clause, the derivation must have entered the YP cycles, for both embedded and matrix clauses. Since one cannot “go back” to an earlier cycle, XP cannot land lower than matrix YP, deriving the height-locality correlation.

Applying this explanation to Mandarin, YM25 suggest that external object topicalization and VC take place within the CP cycle, which is built after the TP and FP cycles. Therefore, they can cross CPs and land in CPs. In contrast, internal object topicalization occurs within the TP cycle and thus cannot cross CP, which is built later. When CP is built, the TP cycle has passed and topicalization cannot land in TP but only CP. By the same logic, internal VC occurs within the FP cycle and cannot cross either TP or CP.

3. Against Williams Cycle effects in Mandarin. We argue against the existence of the WC in Mandarin Chinese. Our counter-arguments are three-fold.

3.1. **HYPERRAISING IN MANDARIN CHINESE.** First, Mandarin, like Cantonese, permits hyperraising, which appears in two distinct variants. One type, as argued by Lee & Yip (2024), involves hyperraising from CP complements selected by raising attitude verbs that encode indirect evidentiality, as illustrated in (11). The other type, as shown by Chen (2023, 2025b), involves hyperraising from CPs selected by other attitude verbs (encoding direct evidence or underspecified) when these verbs are passivized with *bei*, as in (12).²

(11) Hyperraising with raising attitude verbs

- a. Wo tingshuo/ganjue [_{CP} (shuo) **na chang yu** bu hui ting]. (transitive)
 1SG hearsay/feel.like COMP that CL rain not will stop
 ‘I heard that/ feel like the rain will not stop.’
- b. **Na chang yu** tingshuo/ganjue [_{CP} (shuo) _ bu hui ting]. (hyperraising)
 that CL rain hearsay/feel.like COMP not will stop
 ‘It is heard/felt that the rain will not stop.’ (adapted from Lee & Yip 2024:1529n5)

(12) Hyperraising with passivized attitude verbs

- a. Jingcha renwei/huaiyi/xiangxin [_{CP} (shuo) **Zhangsan** hui mousha Lisi]. (active)
 police think/suspect/believe COMP Zhangsan will murder Lisi
 ‘The police thought/suspected/believed that Zhangsan will murder Lisi’
- b. **Zhangsan** bei (jingcha) renwei/huaiyi/xiangxin [_{CP} (shuo) _ hui mousha Lisi].
 Zhangsan BEI police think/suspect/believe COMP will murder Lisi
 Lit. ‘Zhangsan was thought/suspected/believed that (he) will murder Lisi (by the police).’
 (passive) (adapted from Chen 2025b, ex.101)

In both cases, the movement lands in Spec,TP, despite crossing a finite CP. One of Lee & Yip (2024)’s arguments for distinguishing hyperraising to Spec,TP from topicalization is that certain *wh*-words and quantifiers, such as *henduo* ‘many’ NP and *duoshao* ‘how many’ NP, resist topicalization (cf. (13)), but can undergo hyperraising (cf. (14a)). We extend this argument to the *bei* cases in (14b).

- (13) a. {***Henduo ren/** ^{OK}Zhangsan} (ne), wo juede _ bu hui lai.
 many person Zhangsan TOP 1SG think not will come
 Int.: ‘Many people/Zhangsan, I think won’t come.’
- b. {***Duoshao ren/** ^{OK}na-xie ren} (ne), ni juede _ bu hui lai?
 how.many person which-CL.PL person TOP 2SG think not will come
 Int.: ‘How many people/which people are such that you think (they) won’t come?’

² Ura (1994:66,72ff), building on Li (1990), proposes that epistemic modals in Mandarin select finite clauses whose embedded subjects undergo hyperraising into the matrix clause. This account adopts a modal-as-verb analysis (e.g., Lin & Tang 1995). By contrast, Tsai (2015) argues that Chinese modals are functional heads, with epistemic modals positioned above TP, and that the apparent “subjects” preceding them are in fact topicalized. See Hu & Chen (2022) for arguments in favor of analyzing this topicalization as (pure) $\bar{A}$ -movement (but see Chou 2026:17,22 for a more nuanced perspective).

- (14) a. **Henduo ren** tingshuo/ganjue [_{CP} (shuo) _ bu hui lai].
 Many person hearsay/feel COMP not will come
 ‘It is heard/felt that many people will come.’ (see Lee & Yip 2024:1548 for Cantonese)
- b. **Duoshao ren** bei (jingcha) renwei/huaiyi/xiangxin [_{CP} (shuo) _ hui mousha Lisi]?
 how.many person BEI police think/suspect/believe COMP will murder Lisi
 ‘How many people are thought/suspected/believed that (they) will murder Lisi (by the police)?’

Anticipating more discussion on the distribution of probes below, we briefly summarize Lee & Yip (2024)’s and Chen (2025b)’s proposals, which are somewhat different. For Lee and Yip, hyperraising involves direct movement from the embedded Spec,TP to the matrix Spec,TP without an intermediate step through the embedded Spec,CP, which is enabled by phase deactivation via Agree in (indirect) evidential features, as illustrated in (15a) (with intermediate movement to vP omitted). For Chen, hyperraising is enabled when the embedded C hosts a pure [A] (e.g., [ϕ] probe (in addition to a pure [$\bar{A}$] probe, under the assumption that they probe separately and independently, cf. Lohninger et al. 2022; Lohninger & Yip 2023’s independent probing),³ which triggers A-movement to the embedded Spec,CP and enables subsequent A-movement to the matrix Spec,TP, as illustrated in (15b) (with intermediate movement to VoiceP headed by *bei* omitted).

- (15) a. [_{CP} C [_{TP} **Subj** T [ϕ] [_{vP} v+V_[Evid] [_{CP}_[Evid] C_[$\bar{A}$] [_{TP} T ...
- ↑
A-movement
- b. [_{CP} C [_{TP} **Subj** T [ϕ] [_{VoiceP} *bei* V [_{CP} C [ϕ] [$\bar{A}$] [_{TP} T ...
- ↑
A-movement
- ↑
A-mvt.

Our main point is that the WC proposed by YM25, as recapitulated in (16) below, fails to accommodate hyperraising, even if one reformulates it. Specifically, one cannot weaken (16) by relaxing only (i) (Greeson & Sundaesan 2025), since, as proposed by Chen, *bei* hyperraising does proceed from Spec,CP (cf. Mongolian hyperraising, Fong 2019); nor by relaxing only (ii) (Egressy 2026a), since, as proposed by Lee & Yip, evidential hyperraising does cross CP (cf. Zulu hyperraising, Halpert 2019).⁴

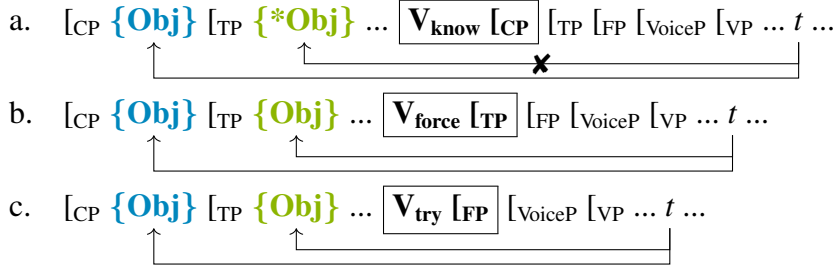
- (16) Williams Cycle: Movement to Spec,XP cannot (i) **proceed from Spec,YP** or (ii) **cross YP**, where Y is higher than X in the functional sequence.

3.2. OBJECT TOPICALIZATION AND DISTRIBUTION OF A/ $\bar{A}$ -PROBES. Second, we emphasize that YM25 overlook that external and internal object topicalizations are different movement types. Recall that there is an apparent height-locality correlation in (17) established in previous literature (see Chen 2023 and references therein), which YM25 take to support the WC.

³ Presumably, verbs like *renwei* ‘think’, which lack an (indirect) evidential feature and allow hyperraising only in the presence of a passive Voice *bei*, select a C with [ϕ] only when they are passivized.

⁴ To our knowledge, only Keine (2020)’s approach has sufficient flexibility to accommodate hyperraising, by allowing a probe to lack a Horizon. Meadows (2024:§6) has an interesting alternative to weaken the WC by exempting “semantically-driven hyperraising”. We will not adjudicate whether these two cases in Mandarin can be reduced to semantics, but we note that this alternative still does not account for the counter-examples in VC (see §3.3), which is argued by YM25 to be triggered by syntactic features.

(17) Object topicalization across (a) CP, (b) TP, and (c) FP boundaries



YM25's approach, however, leaves the other movement properties unaccounted for. For example, external but not internal topicalization allows reconstruction for Principle A, as in (18).

(18) External and internal object topicalization differ in Principle A reconstruction

- a. **Ta-ziji_{i/j}-de erzi** Zhangsan_i dasuan/changshi [PRO_i bipo/mingling Lisi_j [PRO_j 3SG-self's son Zhangsan plan/try force/order Lisi ma _ yidun]].
scold severely
'His_{i/j} son, Zhangsan_i planned/tried to force/order Lisi_j to scold (him) severely.'
- b. Zhangsan_i **ta-ziji_{i/*j}-de erzi** dasuan/changshi [PRO_i bipo/mingling Lisi_j [PRO_j Zhangsan 3SG-self's son plan/try force/order Lisi ma _ yidun]].
scold severely
'Zhangsan_i, his_{i/*j} son, planned/tried to force/order Lisi_j to scold (him) severely.'

Some languages like Dutch is unlike English in allowing Principle A reconstruction only in $\bar{A}$ -movement and not A-movement (Neeleman & van de Koot 2010). Chen (2023, 2025a) suggests that Mandarin is like Dutch in that the lack of Principle A reconstruction is an A-property.

More generally, Mandarin external topicalization exhibits $\bar{A}$ -properties, whereas internal topicalization exhibits a mix of A- and $\bar{A}$ -properties: both external and internal topicalization are not subject to A-Minimality and can cross another DP (like $\bar{A}$ -movement), but internal topicalization lacks weak crossover and reconstruction effects (like A-movement), as summarized in Table 1.⁵

Our view, corroborating Chen (2023, 2025a), is that external and internal topicalizations are triggered by different probes—pure $[\bar{A}]$ vs. composite $[A/\bar{A}]$ —which have different distributions (cf. van Urk 2015; Fong 2019; Lohninger et al. 2022; Lohninger 2025 for a featural typology). More specifically, internal topicalization is triggered by a composite probe $[\phi + \text{TOP}]$ on the internal Top head, as illustrated in (19a), whereas external topicalization is triggered by a pure $\bar{A}$ probe $[\text{TOP}]$ on the external Top head, as shown in (19b).

⁵ Chen (2025a) shows that external topicalization may or may not exhibit weak crossover and reconstruction effects, depending on the height of the co-referring pronoun: such effects arise when the pronoun is above VoiceP, but can be obviated when the pronoun is below VoiceP. She accounts for this by proposing that external topicalization, like internal topicalization, may involve intermediate composite $A/\bar{A}$ -movement to VoiceP, triggered by a composite $[A/\bar{A}]$ probe on the Voice head, but that for external topicalization the terminating movement to the external TopP is purely $\bar{A}$ -movement.

Properties	A	$\bar{A}$	Dinka, English, Mandarin composite A/ $\bar{A}$ -movement	Mandarin internal topicalization	Mandarin external topicalization
New antecedents for anaphor binding	✓	*	✓	✓	*
No weak crossover	✓	*	✓	✓	*
No reconstruction for Principle C	✓	*	✓	✓	*
Long-distance ...	*	✓	✓	✓	✓
... crossing c-commanding NPs					
Long-distance ...	*	✓	✓ (D)	*	✓
... crossing finite CPs			* (E, M)		

Table 1. A/ $\bar{A}$ -properties of external and internal topicalization in Mandarin (Chen 2025a)

- (19) a. Internal object topicalization as successive-cyclic composite A/ $\bar{A}$ -movement

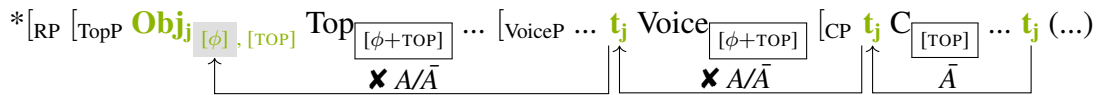
$$[\text{TP Subj}_i[\phi] \text{ T}_{[\phi]} [\text{TopP } \text{Obj}_j[\phi], [\text{TOP}] \text{ Top}_{[\phi+\text{TOP}]} \dots [\text{VoiceP } t_i \text{ t}_j \text{ Voice}_{[\phi+\text{TOP}]} [\dots \text{ t}_j (\dots)]$$

b. External object topicalization via intermediate composite A $\bar{A}$ -movement

$$[\text{TopP } \text{Obj}_j[\phi], [\text{TOP}] \text{ Top}_{[\text{TOP}]} [\text{TP Subj}_i[\phi] \text{ T}_{[\phi]} \dots [\text{VoiceP } t_i \text{ t}_j \text{ Voice}_{[\phi+\text{TOP}]} [\dots \text{ t}_j (\dots)]$$

Furthermore, locality is constrained by the distribution of probes. In Mandarin, Voice heads host composite $[A/\bar{A}]$ probes, whereas C heads host pure $[\bar{A}]$ probes (see Chen 2023, 2025a,b).⁶ Consequently, internal object topicalization across CP is ruled out due to the familiar Ban on Improper Movement: $\bar{A}$ -movement renders the A-features of moved elements invisible, blocking further composite A/ $\bar{A}$ -movement, as in (20). Importantly, the invisibility of A-features of purely $\bar{A}$ -moved elements is evident from the so-called $\bar{A}$ -opacity effects (Obata & Epstein 2011)—the absence of ϕ -agreement and A-Minimality effects on further A-movement—as well as the correlated weak crossover and reconstruction behaviors.

- (20) Improper composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement



Note that one cannot simply supplement a correlation between height and A/ $\bar{A}$ -properties, for example, that movement to the TP domain is always A-like, given that Georgian TP-internal *wh*-movement exhibits pure $\bar{A}$ -properties rather than composite A/ $\bar{A}$ -properties (Bondarenko 2024). The A/ $\bar{A}$ -distinction is thus independently needed, rendering the WC superfluous, at least for Mandarin.

Before proceeding, we also note that internal topicalization can land below TP while crossing a TP. As illustrated in (21), object topicalization may land below deontic modals, as noted by Chen & Yip (2025). Deontic modals project lower than T (Tsai 2015; Yip & Lee 2022), as also assumed by YM25 ($T \succ M^{Deo}$). Yet, the object can still cross a TP taken by ‘force’-type verbs, as in (22). This constitutes yet another empirical challenge to the height-locality correlation.

⁶ And pure $[A]$ probes when taken by passivized verbs (see footnote 3), which probe separately and independently.

- (21) Lisi yinggai [ZHE-jian shi]_i haohao diaocha _i.
 Lisi should this-CL matter well investigate
 Lit.: ‘Lisi should, this matter (but not that matter), investigate (it) thoroughly.’
- (22) Lisi yinggai [ZHE-jian shi]_i bipo/mingling Zhangsan_j [PRO_j haohao diaocha _i].
 Lisi should this-CL matter force/command Zhangsan well investigate
 Lit.: ‘Lisi should, this matter (but not that matter), force/command Zhangsan to investigate (it) thoroughly.’

3.3. VERB COPYING CONSTRUCTIONS AND SUBJECT IDENTITY. We now turn to Verb Copying constructions. VC arises when the same verb combines with a post-verbal object and a second post-verbal phrase, such as a resultative or degree phrase headed by *de* (Gouguet 2006; Cheng 2007, among others), as illustrated in (23a). The higher verb with the object (VP2) is typically analyzed as an adjunct (Huang 1982, 1992; Cheng 2007), and verb copying can be derived via sideward movement (Cheng 2007; cf. Nunes 2004), as in (24).⁷

- (23) a. Lisi hui [VP1 [VP2 **kan** zhe-ben shu] **kan-de** [DEP *de* hen kuai]].
 Lisi will read this-CL book read-DE very fast
 ‘Lisi will reading this book read (it) fast.’
- b. {[VP2 **kan** zhe-ben shu]} Lisi {[VP2 **kan** zhe-ben shu]} hui [VP1 *t*_{VP2} **kan-de** [DEP *de* hen kuai]].
 read this-CL book Lisi read this-CL book will read-DE
 very fast
 ‘Lisi reading this book will read (it) fast.’

(24) VP copying via sideward movement (adapted from Cheng 2007:160)

- a. [VP V [DEP ...]] (Baseline)
- b. [VP V [DEP ...]] V (Sideward movement)
 └──────────┬──────────
 Copy ↑
- c. [VP1 V [DEP ...]] [VP2 V <Merge> DP/bare noun] (Merging objects)
- d. [VP1 [VP2 V DP/bare noun] [VP1 V [DEP ...]]] (Adjunction)
- e. (Subj) [VP2 V DP/bare noun] (Subj) ... [VP1 *t*_{VP2} [VP1 V [DEP ...]]]
 ↑ Move (Optional: Further movement in (23b))

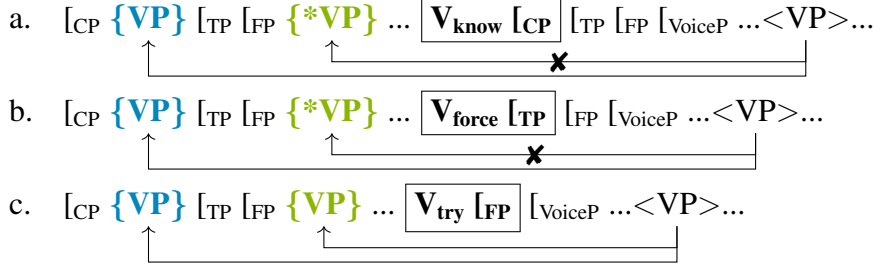
YM25 generalize the locality profiles of external VC and internal VC in (25), assuming that internal VC lands in FP but not higher.⁸

⁷ YM25 derive the copying effects differently (cf. Meadows & Yan 2023), but the choice does not bear on the WC.

⁸ This is already at odds with the internal VP2 above modals in (23b).

(25) VC constructions across (a) CP, (b) TP, and (c) FP boundaries

(to be challenged)



Crucial to their analysis is that ‘try’-type verbs select an FP, which is smaller than TP. However, upon examining their classification in (26), we note that it does not align with Huang (2022:24)’s classification of verbs based on complementation types, which they cite as its source.

(26) YM25’s verb classification (their ex.24)

- a. ‘Know’-type: *faxian* ‘find out’, *fouren* ‘deny’, *juede* ‘feel’, *renwei* ‘think’, *zhidao* ‘know’
- b. ‘Force’-type: *bi(po)* ‘force’, *qiangpo* ‘force’, *quan* ‘urge, persuade’, *yinyou* ‘lure’, *zhidao* ‘instruct’
- c. ‘Try’-type: *changshi* ‘try’, *dasuan* ‘intend’, *jihua* ‘plan’, *jueding* ‘decide’, *kaishi* ‘begin’

In particular, Huang (2022) argues that *dasuan* ‘plan’ (also *jihua* ‘plan’ and *jueding* ‘decide’) selects at least a TP-sized “situation” complement (cf. Wurmbrand & Lohninger 2023) and can host future tense markers such as *yao* or *jiang* (N. Huang 2015), patterning with ‘force’-type verbs like *bipo* ‘force’. VC can cross the TP with *yao* as in (27).

- (27) Zhangsan {*kan zhe-ben shu*} *dasuan* [TP PRO_i *yao kan-de hen kuai*].
 Zhangsan read this-CL book plan FUT read-DE very fast
 ‘Zhangsan {reading this book} planned to read (it) fast.’

We suggest that ‘force’-type and ‘try’-type verbs differ not in the size of their complements, but rather whether they involve object control or subject control, which raises the question of why control type should affect the locality of VC.

We further suggest that *subject identity* is the crucial factor. In object control, the matrix subject and the embedded PRO need not be co-referential, whereas in subject control predicates, the matrix subject and the embedded PRO must be co-referential. Evidence comes from cases where the matrix object is a reflexive and thus co-referential with the subject and embedded PRO. In (28), VC can cross the complement of ‘force’-type verbs, contrasting (10b) where internal VC is not possible with the matrix object *Lisi*. More strikingly, when subject identity holds, such movement can even cross a finite CP with a bound pronoun subject, as in (29). This pattern falsifies the generalization in (25).

- (28) {*Kan zhe-ben shu*} Zhangsan_i {*kan zhe-ben shu*} *bipo/mingling ta-ziji* [PRO_i
 read this-CL book Zhangsan read this-CL book force/order 3SG-self
kan-de hen kuai.
 read-DE very fast
 ‘{Reading this book}, Zhangsan {reading this book} forced/ordered himself to read (it) fast.’

- (29) {**Kan zhe-ben shu**} Zhangsan_i {**kan zhe-ben shu**} xiangxin/zhidao [_{CP} **ta_{i/*j}** hui
 read this-CL book Zhangsan read this-CL book believe/know 3SG will
 kan-de hen kuai].
 read-DE very fast
 ‘{Reading this book}, Zhangsan {reading this book} believes/knows that he will read (it) fast.’

We argue that the possibility of internal VC is constrained not by locality, but rather by the requirement in (30):

- (30) Subject identity in Verb Copying constructions
 The understood subjects of the copied verb phrase (“VP2”) and the in-situ verb phrase (“VP1”) must be co-referential.

This requirement finds its roots in Huang (1993)’s observation that reflexive pronouns within a fronted VP are obligatorily bound by the embedded subject, contrasting with topicalization of just the reflexive, as shown in (31). To capture this, Huang (1993) proposes that the fronted constituent is not a bare VP but rather a VoiceP (vP in his terms) containing a trace of the embedded subject, which binds the reflexive pronoun.

- (31) a. **Ta-ziji_{i/j}** Zhangsan_i xiangxin/zhidao [_{CP} Lisi_j ma-guo _].
 3SG-self Zhangsan believe/know Lisi scold-EXP
 ‘Himself_{i/j}, Zhangsan_i believes/knows that Lisi_j scolded (him_i/himself_j)’
 b. [_{VoiceP} **t_j Ma ta-ziji_{j/*i}**] Zhangsan_i xiangxin/zhidao [_{CP} **Lisi_j** bu hui _].
 scold 3SG-self Zhangsan believe/know Lisi NEG will
 ‘Scold himself_{j/*i}, Zhangsan_i believes/knows that Lisi_j will not (do it).’

Along the same lines, we propose that what is copied in VC is not a bare VP but a VoiceP that includes its own subject, which we take to be PRO, as illustrated in (32).

- (32) a. [_{VoiceP} **PRO_i Kan zhe-ben shu**] Zhangsan_i xiangxin/zhidao [_{CP} **ta_i** hui [_{VoiceP} **t_i**
 read this-CL book Zhangsan believe/know 3SG will
 [_{VP} kan-de hen kuai]]].
 read-DE very fast
 ‘Reading this book, Zhangsan believes/knows that he will read (it) fast.’
 b. Zhangsan_i [_{VoiceP} **PRO_i kan zhe-ben shu**] xiangxin/zhidao [_{CP} **ta_i** hui [_{VoiceP} **t_i**
 Zhangsan read this-CL book believe/know 3SG will
 [_{VP} kan-de hen kuai]]].
 read-DE very fast
 ‘Zhangsan reading this book believes/knows that he will read (it) fast.’

We further propose that external VC lands on the external topic position, whereas internal VC lands on the internal topic position, probed by pure [$\bar{A}$] and composite [$A/\bar{A}$] features, respectively. As internal (object) topicalization resists reconstruction, unlike external (object) topicalization, we expect a parallel contrast in VC: external VC allows the copied VoiceP to reconstruct to its base position and be bound by the embedded subject, whereas internal VC does not

permit reconstruction. Consequently, the PRO in the copied VoiceP can only be controlled by the closest DP, namely the matrix subject. If the matrix subject is not co-referential with the embedded subject, the two VoicePs receive distinct understood subjects, violating (30). These patterns are schematized in (33). In cases where internal VC is grammatical, the required co-reference is ensured in configurations of subject control, object control with reflexive objects, or finite CP with bound pronoun subjects.⁹

- (33) a. $[_{TopP} \text{ [VoiceP PRO}_k \text{ V2-Obj]}_{[\phi+TOP]} \text{ Top}_{[TOP]} [_{TP} \text{ Subj}_i \dots [_{CP/TP} \text{ Subj}_k \dots [_{VoiceP} \boxed{t} \text{ V1 deP}]] \dots]$
↑
reconstruction
- b. $[_{TP} \text{ Subj}_i [_{TopP} \text{ [VoiceP PRO}_{*k/\#i} \text{ V2-Obj]}_{[\phi+TOP]} \text{ Top}_{[\phi+TOP]} \dots [_{CP/TP} \text{ Subj}_k \dots [_{VoiceP} \boxed{t} \text{ V1 deP}]] \dots]$
↑
control *no reconstruction*

In sum, YM25’s conclusion is based on an incomplete generalization. Once subject identity is taken into account, VC does not exhibit WC effects and in fact provides evidence against the applicability of the WC in Mandarin Chinese.

4. Conclusion. To conclude, we have refuted YM25’s arguments for the Williams Cycle in Mandarin. First, Mandarin allows hyperraising, which directly contradicts the WC. Second, we show that external and internal object topicalizations show different A/ $\bar{A}$ -movement properties, a distinction that the WC fails to capture. In contrast, an account with independently diagnosable distribution of pure $\bar{A}$ -probes and composite A/ $\bar{A}$ -probes captures the movement properties all at once, including the locality profiles. Additionally, we show that internal topicalization can land below TP while crossing a TP. Third, we identify a previously unnoticed requirement in Verb Copying constructions: the understood subjects of the copied and in-situ VPs must be co-referential. Once this subject identity requirement is satisfied, YM25’s generalization breaks down—internal VC can cross CP and TP, in addition to the smaller FP.

The inapplicability of the WC in Mandarin has an implication beyond the language, namely, that the WC cannot be universal. Most previous work of the WC focuses on how movement dependencies within a single language conform to it, and rarely discusses the role of cross-linguistic variations. We believe that capturing languages that do not exhibit the WC effects is as important as capturing those that do. Any successful theory of language should leave room for variation (ideally, of course, with proper restriction). Variation may inform the theory itself, too. In YM25’s Level Embedding approach, for example, it would be hard to conceive that some languages build structures in parallel workspaces whereas Mandarin does not, a variation on the architectural level. On the other hand, theories that attribute variation to features on probes are conceptually more plausible, such as the featural approach to A/ $\bar{A}$ -distinction (e.g., Chen 2023; Lohninger 2025) as well as the Horizon approach (Keine 2020). We hope that the discussion here provides a useful basis for further comparison between different theories in this domain.

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⁹ We further assume that the copied VP does not pass through Spec,CP as an intermediate landing site, instead deriving its position via sideward movement. We leave a detailed implementation of this analysis for future work.

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